Beginner-level Java programming problems

Module 1: Input, Output and Basic Arithmetic

Problem 1: Student Registration System

Scenario: A university admission portal collects a student's registration number before generating an application form.

Problem Statement:
Develop a Java program that accepts an integer from the user and displays the entered value along with a confirmation message.

Concepts Covered

- Standard Input
- Standard Output
- Scanner Class

Problem 2: Product Price Calculator

Scenario: A supermarket needs to calculate the total cost of two products having decimal prices.

Problem Statement
Develop a Java program that accepts two floating-point numbers representing product prices and calculates their total cost.

Concepts Covered

- Floating-point arithmetic
- User Input

Problem 3: Digital Wallet Balance Exchange

Scenario: Two customers accidentally transfer money into each other's digital wallets.

Problem Statement
Develop a Java application that swaps two integer values without losing any data and displays the values before and after swapping.

Concepts Covered

- Variables
- Swapping Logic

Problem 4: Scientific Calculator

Scenario: An engineering calculator performs arithmetic operations on complex numbers.

Problem Statement
Develop a Java program that accepts two complex numbers and displays their sum.

Concepts Covered

- Object Representation
- Arithmetic Operations

Module 2: Decision Making

Problem 5: Online Voting Eligibility

Scenario
An election portal verifies whether a citizen is eligible to vote.

Problem Statement
Develop a Java program to determine whether an entered age satisfies the voting eligibility criteria.

Problem 6: Electricity Billing System

Scenario
An electricity board classifies meter readings as positive or invalid.

Problem Statement
Develop a Java program that determines whether a given number is positive, negative, or zero.

Problem 7: Banking Transaction Validation

Scenario
A banking system processes transactions only if the entered amount is even.

Problem Statement
Determine whether a given integer is even or odd.

Problem 8: Student Performance Analysis

Scenario
Three students participate in a coding competition.

Problem Statement
Develop a Java application that identifies the student with the highest score among three entered marks.

Problem 9: Calendar Utility

Scenario
A calendar application displays weekday names and validates dates.

Problem Statement
Develop Java programs to:

- Display the weekday for a given number (1–7).
- Determine the number of days in a given month and year.
- Check whether a given year is a leap year.

Module 3: Loops and Number Processing

Problem 10: Prime Number Generator

Scenario
A cybersecurity application generates prime numbers for encryption algorithms.

Problem Statement
Generate all prime numbers between **1 and N**.

Problem 11: Mathematical Number Analyzer

Scenario
A mathematics learning application helps students explore different number properties.

Problem Statement
Develop separate Java programs to determine whether a given number is:

- Palindrome
- Armstrong Number
- Neon Number
- Perfect Square

Problem 12: Fibonacci Energy Model

Scenario
A simulation software generates Fibonacci values for computational modelling.

Problem Statement
Generate the Fibonacci series up to **N** terms and calculate the sum of all even Fibonacci numbers.

Problem 13: Financial Calculator

Scenario
A bank provides customers with loan estimation tools.

Problem Statement
Develop Java programs to calculate:

- Simple Interest
- Compound Interest

Problem 14: Mathematics Utility

Scenario
A scientific calculator supports advanced mathematical operations.

Problem Statement
Develop separate Java programs to calculate:

- Factorial
- GCD
- LCM
- Power of a Number
- Permutations
- Combinations

Module 4: Number System Conversion

Problem 15: Digital Electronics Toolkit

Scenario
An embedded systems engineer frequently converts numbers between different number systems.

Problem Statement
Develop Java programs to perform the following conversions:

- Decimal to Binary/Octal/Hexadecimal
- Binary/Octal/Hexadecimal to Decimal
- Binary to Octal
- Octal to Binary

Problem 16: Number System Calculator

Scenario
A processor performs arithmetic directly on binary and octal numbers.

Problem Statement
Develop Java applications to perform:

- Addition
- Subtraction
- Multiplication

for numbers belonging to different number systems.

Module 5: Mathematical Computations

Problem 17: Geometry Calculator

Scenario
An architectural design application calculates dimensions of different structures.

Problem Statement
Develop Java programs to calculate:

- Area and perimeter of a circle
- Perimeter of a rectangle

Problem 18: Statistics Calculator

Scenario
A data analyst wants to summarize collected observations.

Problem Statement
Develop Java programs to:

- Calculate the average of three numbers.
- Calculate the sum of digits of an integer.
- Count the number of digits.
- Print every digit individually.
- Extract a digit from a specified position.

Module 6: Character Processing

Problem 19: Text Processing Utility

Scenario
A text editor performs character-based operations during document processing.

Problem Statement
Develop Java programs to:

- Display ASCII values.
- Determine whether a character is a vowel or consonant.

Module 7: Pattern Generation using Nested Loops

Learning Objective

Develop Java programs using nested loops to generate various console-based patterns. These exercises strengthen students' understanding of loop constructs, conditional statements, and output formatting.

Category A: Square and Rectangle Patterns

Problem 20: Employee Attendance Grid

Scenario: An organization wants to display employee attendance in a tabular grid format.

Problem Statement:
Develop a Java program to generate the following **solid square pattern** using the '*' symbol for a given value of **N**.

Example (N = 5)

```
*****
*****
*****
*****
```

```
*****
```

Problem 21: Event Seating Layout

Scenario: An event organizer wants to visualize a rectangular seating arrangement.

Problem Statement:
Develop a Java program to print a rectangular star pattern based on the specified number of rows and columns.

Example

```
******
******
******
```

Category B: Half Pyramid Patterns

Problem 22: Staircase Construction

Scenario: An architect wants to display the step-wise construction of a staircase.

Problem Statement:
Develop a Java program to generate a left-aligned half pyramid using the '*' symbol.

```
*
**
***
****
*****
```

Problem 23: Reverse Staircase

Scenario: During demolition planning, the staircase is removed from top to bottom.

Problem Statement:
Develop a Java program to generate an inverted half pyramid.

```
*****
****
***
**
*
```

Category C: Right-Aligned Pyramid Patterns

Problem 24: Building Elevation Design

Scenario: A civil engineer wants to visualize the elevation of a building.

Problem Statement:
Develop a Java program to generate the following right-aligned half pyramid.

```
    *
   **
  ***
 ****
*****
```

Problem 25: Reverse Building Elevation

Scenario: Display the reverse view of the building elevation.

Problem Statement:
Develop a Java program to generate the following pattern.

```
*****
 ****
  ***
   **
    *
```

Category D: Full Pyramid Patterns

Problem 26: Festival Decoration

Scenario: Decorative lights are arranged in the shape of a pyramid.

Problem Statement:
Develop a Java program to print a full pyramid.

```
    *
   ***
  *****
 *******
*********
```

Problem 27: Inverted Decorative Pyramid

Scenario: Display the decorative structure in reverse order.

Problem Statement:
Develop a Java program to print an inverted full pyramid.

```
*********
 *******
  *****
   ***
    *
```

Category E: Diamond Patterns

Problem 28: Diamond Logo Design

Scenario: A graphic designer wants to display a diamond-shaped company logo.

Problem Statement:
Develop a Java program to generate the following diamond pattern.

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

Category F: Hollow Patterns

Problem 29: Window Frame Design

Scenario: A carpenter designs a square window frame where only the border is visible.

Problem Statement:
Develop a Java program to print a hollow square.

```
*****
*   *
*   *
*   *
*****
```

Problem 30: Hollow Pyramid Design

Scenario: A monument architect wants to display only the outer structure of a pyramid.

Problem Statement:
Develop a Java program to generate a hollow pyramid.

```
    *
   * *
  *   *
 *     *
*********
```

Category G: Numeric Patterns

Problem 31: Student Roll Number Display

Scenario: A classroom management system displays roll numbers in increasing order.

Problem Statement:
Develop a Java program to print the following numeric triangle.

```
1
12
123
1234
12345
```

Problem 32: Number Triangle

Scenario: A mathematics learning application displays number patterns for practice.

Problem Statement:
Generate the following pattern.

```
1
22
333
4444
55555
```

Problem 33: Floyd's Triangle

Scenario: An educational application demonstrates sequential number generation.

Problem Statement:
Develop a Java program to generate Floyd's Triangle.

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Problem 34: Pascal's Triangle

Scenario: A combinatorics learning tool displays binomial coefficients using Pascal's Triangle.

Problem Statement:
Develop a Java program to generate Pascal's Triangle for **N** rows.

Example

```
        1
      1   1
    1   2   1
  1   3   3   1
1   4   6   4   1
```

Category H: Alphabet Patterns

Problem 35: Employee Grade Display

Scenario: A company assigns alphabetical grades to employee performance.

Problem Statement:
Develop a Java program to generate the following alphabet pattern.

```
A
AB
ABC
ABCD
ABCDE
```

Problem 36: Character Pyramid

Scenario: A typing tutor application displays alphabetical pyramids for practice.

Problem Statement:
Generate the following pattern.

```
A
BB
CCC
DDDD
EEEEE
```

Category I: Advanced Symmetric Patterns

Problem 37: Hourglass Design

Scenario: A digital art application creates an hourglass-shaped design.

Problem Statement:
Develop a Java program to generate the following pattern.

```
*********
 *******
  *****
   ***
    *
   ***
  *****
 *******
*********
```

Problem 38: Butterfly Pattern

Scenario: A graphic design application generates decorative butterfly-shaped patterns.

Problem Statement:
Develop a Java program to generate the butterfly pattern.

```
*       *
**     **
***   ***
**** ****
*********
**** ****
***   ***
**     **
*       *
```

Problem 39: X Pattern

Scenario: A digital display board shows decorative symbols for visual effects.

Problem Statement:
Develop a Java program to print the following **X-shaped** pattern.

```
*       *
 *     *
  *   *
   * *
    *
   * *
  *   *
 *     *
*       *
```

Category J: Miscellaneous Patterns

Problem 40: Multiplication Table Generator

Scenario: An educational application helps students practice multiplication tables.

Problem Statement:
Develop a Java program that prints the multiplication table of a given number up to a specified limit.

Example

```
5 × 1 = 5
5 × 2 = 10
...
5 × 10 = 50
```

Problem 41: Chessboard Pattern

Scenario: A game development company wants to display a chessboard layout in the console.

Problem Statement:
Develop a Java program to generate the following checkerboard pattern.

```
* * * *
 * * * *
* * * *
 * * * *
```

This categorization is more aligned with **engineering laboratory manuals**. Instead of listing patterns randomly, it organizes them into **10 logical categories**:

1. Square & Rectangle Patterns
2. Half Pyramid Patterns
3. Right-Aligned Patterns
4. Full Pyramid Patterns
5. Diamond Patterns
6. Hollow Patterns
7. Numeric Patterns
8. Alphabet Patterns
9. Advanced Symmetric Patterns
10. Miscellaneous Patterns

This progression helps students gradually master **nested loops**, **spacing logic**, and **pattern design techniques**, while each exercise is presented in a practical scenario rather than as a simple "print this pattern" task.

Module 8: Integrated Mini Applications

Problem 21: Multiplication Table Generator

Develop a multiplication table generator for any entered integer.

Problem 22: Binary Calculator

Develop a calculator capable of adding two binary numbers.

Problem 23: Mirror Number Generator

Develop a Java application that determines the mirror inverse of a given number.

Problem 24: Number Classification Suite

Develop a menu-driven Java application that allows the user to select different mathematical operations such as:

- Prime Check
- Armstrong Check
- Palindrome Check
- Fibonacci Generation
- Factorial
- GCD & LCM
- Number Conversion